



May 2024

(BT)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.

❖ *Linear Graph should be provided.*

20 Marks

- The unit of reaction rate is written as
 A) mol/L
 B) L.s/mol
 C) mol/L.s
 D) L/mol.s
- The stoichiometry of a reaction is $2\text{NO}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{N}_2\text{O}_5$. The rate of formation of N_2O_5 is given by
 A) $r_{\text{N}_2\text{O}_5} = -r_{\text{NO}_2}/2$
 B) $r_{\text{N}_2\text{O}_5} = -r_{\text{NO}_2}$
 C) $r_{\text{N}_2\text{O}_5} = 2(-r_{\text{NO}_2})$
 D) $r_{\text{N}_2\text{O}_5} = -r_{\text{NO}_2}/4$
- The unit of reaction rate constant is determined by
 A) (Concentration) $^{n-1}$. Time $^{-1}$
 B) (Concentration) $^{n-1}$. Time $^{1-n}$
 C) (Concentration) $^{1-n}$. Time
 D) (Concentration) $^{1-n}$. Time $^{-1}$
- On doubling the concentration of reactant, the rate of reaction triples. Calculate the reaction order.
 A) 1
 B) 1.6
 C) 2
 D) 2.7
- If chemical potential of reaction $\text{A} \rightarrow \text{B}$ is same i.e; $\mu_{\text{A}} = \mu_{\text{B}}$, then reaction _____.
 A) Proceeds forward
 B) At equilibrium
 C) Proceeds backward
 D) is irreversible
- Standard heat of reaction is _____ for exothermic reaction.
 A) Negative
 B) Positive
 C) Zero
 D) Either positive or negative
- Gibb's free energy is given by
 A) $\Delta G = \Delta H + T\Delta S$
 B) $\Delta G = \Delta H + \Delta TS$
 C) $\Delta G = \Delta H - \Delta TS$
 D) $\Delta G = \Delta H - T\Delta S$
- The value of Gibb's free energy at equilibrium
 A) $\Delta G > 0$
 B) $\Delta G < 0$
 C) $\Delta G = 0$
 D) $\Delta G = 1$
- _____ method is suitable to interpret the scattered kinetic data
 A) Integral
 B) Half-life
 C) Differential
 D) Initial Rate
- The fractional volume is defined as _____
 A) $\frac{V_{\text{at } X=1} - V_{\text{at } X=0}}{V_{\text{at } X=0}}$
 B) $\frac{V_{\text{at } X=1} - V_{\text{at } X=0}}{V_{\text{at } X=1}}$
 C) $\frac{V_{\text{at } X=0} - V_{\text{at } X=1}}{V_{\text{at } X=0}}$
 D) $\frac{V_{\text{at } X=0} - V_{\text{at } X=1}}{V_{\text{at } X=1}}$
- Kinetic expression determined by Initial rate method involves
 A) Measuring initial concentration
 B) Measuring the initial reaction rate
 C) Measuring initial fractional volume change
 D) Measuring initial conversion

12. The testing kinetics of new reaction is carried out in _____ Reactor
 A) Semi batch B) Plug flow
 C) Mixed flow D) Batch
13. Time required to process one reactor volume of feed is
 A) Space time B) Retention time
 C) Residence time D) Treatment time
14. The performance equation for CSTR with partially converted reactant is
 A) $t = \frac{(X_{Af} - X_{Ai})}{-r_A}$ B) $t = C_{A0} \frac{X_{Ai}}{-r_A}$
 C) $t = C_{A0} \frac{(X_{Af} - X_{Ai})}{-r_A}$ D) $t = C_{A0} \frac{X_{Af}}{-r_A}$
15. PFR connected in series can be simplified by _____
 A) Diverting the molar feed rate B) Adding the individual reactor volume in series chain
 C) Making volumetric flow rate same in each branch D) deleting equal volume reactor in series chain
16. For equal size CSTRs in series carrying out 1st order reaction, the concentration at the outlet of nth reactor is _____
 A) $C_N = \frac{C_0}{(\tau k + 1)^N}$ B) $C_N = \frac{C_0}{(\tau + 1)^N}$
 C) $C_N = \frac{C_0}{(\tau k - 1)^N}$ D) $C_N = \frac{C_0}{(\tau k + 1)}$
17. If M is the total amount of tracer injected and v is the volumetric flow rate of the effluent, then what is the relationship between cumulative and exit age distribution?
 A) $E = \frac{M}{v}$ B) $E = \frac{v}{M}$
 C) $E = \frac{C}{\int_0^\theta C dt}$ D) $E = \frac{C}{\int_0^\theta M dt}$
18. The mean residence time obtained from cumulative age distribution is _____
 A) $\bar{t} = \int_0^\theta t E dt$ B) $\bar{t} = \int_0^\theta t C dt$
 C) $\bar{t} = \int_0^\theta E dt$ D) $\bar{t} = \int_0^\theta C dt$
19. The exit age distribution of fluid leaving a vessel is used
 A) to know the activation energies of a reaction B) to know the reaction rate constants
 C) to study the reaction mechanism D) to study the non-ideal flow in the vessel
20. RTD in MFR is determined by
 A) $E = \frac{e^{t/\tau}}{\tau}$ B) $E = \frac{e^{-t/\tau}}{\tau}$
 C) $E = \frac{e^{-\tau/t}}{\tau}$ D) $E = \frac{e^{\tau/t}}{\tau}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) How chemical reactions are classified? Explain each one of them with example.	8	L2	1	1
	b) How is feasibility of a chemical reaction determined using Gibb's free energy? Calculate Gibb's free energy and state whether it is spontaneous for a reaction carried at 298K having $\Delta H = -495.2$ kJ and $\Delta S = -136$ J/K.	8	L3	2	2
2.	a) Define: (i) Reaction Rate (ii) Reaction Mechanism (iii) Rate constant (iv) Reaction Order.	8	L2	1	1
	b) Derive Van't Hoff's equation and calculate the equilibrium constant at 298K for the reaction. $N_2O_4 \rightarrow 2NO_2$ Standard free energy of formation at 298K are 97540J/mol and 51310J/mol for N_2O_4 & NO_2 respectively.	8	L3	2	2

Unit – II

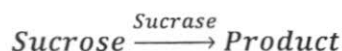
3. a) Differentiate between: (i) Elementary and non-elementary reaction,
(ii) Molecularity and order of reaction.
- b) In studying the kinetics of decomposition of reaction, the concentration of reactant is determined analytically at different time. The following results were obtained. Determine order and rate constant.

Time, min	0	10	20	40	100	125
C, mol/L	0.1	0.0714	0.0556	0.0385	0.02	0.0167

4. a) Explain collision and transition state theory. How temperature dependent terms are calculated from these theories?
- b) The partial pressure of reactant A was observed as a function of time at 600K with the result given below. Calculate the order and rate constant.

Time, s	0	1000	2000	3000	4000
Partial pressure, torr	820	572	399	278	194

5. a) A common rule of temperature is that the rate of reaction doubles for each 10°C rise in temperature. What activation energy would this suggest at a temperature of 25°C?
- b) Sucrose is hydrolyzed at room temperature by the catalytic action of the enzyme sucrase as



Following kinetic data are obtained in a batch reactor by starting with a sucrose concentration of $C_{A0}=1.0\text{mmol/L}$ and enzyme concentration $C_{E0}=0.01\text{mmol/L}$.

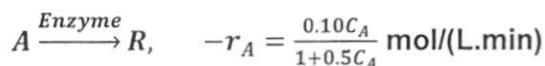
C_A , mmol/L	0.68	0.38	0.16	0.04	0.006
Time, h	2	4	6	8	10

Obtain a kinetic equation and its constants for this reaction.

Assume kinetic equation: $-r_A = \frac{kC_A C_{E0}}{(C_A + M)}$

Unit – III

6. a) Derive the time required to reduce the initial reactant concentration to C_A mol/L in batch reactor.
- b) Define RTD (Residence Time Distribution). In detail explain the measurement of RTD in reactor.
7. a) A specific enzyme acts as catalyst in the fermentation of substrate A. At a given enzyme concentration in the aqueous feed stream of 25L/min, determine the size of PFR needed to achieve 95% conversion of reactant A ($C_{A0}=2\text{mol/L}$). The kinetic and stoichiometry of the fermentation reactions are given by.



- b) What is a tracer? List and explain the characteristics of a good tracer.
8. a) The elementary liquid phase reaction $A+B \rightarrow R+S$ is conducted in a setup consisting of a MFR (Mixed Flow Reactor) into which 2 reactant streams are introduced followed by a PFR (Plug Flow Reactor). The reactant B is used in large excess so that reaction is first order with respect to A. If the reactor combination is reversed by connecting PFR first followed by MFR, how would this change affect conversion?
- b) What is C, E and F curves? Explain the significance of C, E and F curves in determining the degree of non-ideality in reactor.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
